

MD IRFAN

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PROFESSIONAL SUMMARY

[Profile Video](#)

Software Developer with 3+ years of experience building scalable backend systems using Java, Spring Boot, and microservices in a core banking environment supporting 24,000+ branches and 5,000+ daily transactions. Proven in improving performance and reliability through event-driven architectures and optimized APIs. Experienced in full-stack development with React and Angular, with strong DSA (650+ problems) and system design, and exposure to AI-driven applications using LLMs and RAG.

TECHNICAL SKILLS

- **Languages:** Java (Core, Advanced), JavaScript, TypeScript, SQL, Python.
- **Backend & Frameworks:** Spring Boot, Spring MVC, Microservices, REST APIs, Spring Data JPA, Hibernate, Maven.
- **Frontend:** React (Primary), Angular, HTML5, CSS3, Tailwind CSS.
- **Architecture:** Distributed Systems, Event-Driven, API Gateway, Spring Cloud (Config, Eureka, Gateway), Resilience4j.
- **Databases / Caching & Messaging:** PostgreSQL, MySQL, MongoDB, Redis, RDBMS, Apache Kafka, RabbitMQ.
- **Security & Testing:** Spring Security, JWT, OAuth2, OpenID Connect, JUnit, Mockito.
- **Cloud & DevOps:** AWS (EC2, S3, RDS, IAM), Docker, Kubernetes, CI/CD (GitHub Actions), Git, Postman.
- **AI / ML & CS Fundamentals:** LLM, RAG, NLP, Scikit-learn, Pandas, NumPy, DSA, OOP (SOLID), Design Patterns, Agile/Scrum, SDLC.

EXPERIENCE

Tata Consultancy Services | System Engineer

June 2023 – Present

- Worked on core banking transactions like **Branch Creation** and **CCPC Inward Clearing** in a live production system supporting **24,000+ branches** and **2 lakh+ users** worldwide, handling **5000+ transactions daily**.
- Built and improved a **Kafka-based** flow to process cheque data asynchronously, reducing processing time by **~30%** and enabling smooth integration with **ML models** for validation and fraud checks.
- Developed **REST APIs** using **Spring Boot** in **microservices** architecture with proper validations, exception handling, and secure transaction logic, supporting high concurrent users with stable performance.
- Implemented **Maker–Checker** functionality to ensure transactions are approved before saving, reducing incorrect entries and improving audit compliance by **~40%**.
- Improved performance by optimizing **SQL queries** and backend logic, reducing API response time by **~20%** in high-load scenarios.
- Led a team of **5 developers** in **Agile setup**, handling sprint planning and production issues, and reduced bug resolution time by **~25%** through better coordination and mentoring.
- Worked on frontend transaction screens with **dynamic forms, validations, and API integration**, reducing manual errors and improving user efficiency by **~30%**.

PERSONAL PROJECTS

Codexa AI (Lovable Inspired) | Ready to Use App Builder

[Demo](#) | [Code](#) | [Link](#)

- Designed and developed an AI-powered full-stack application generator that converts natural language prompts into production-ready applications using **Spring Boot, React, and LLMs**, reducing manual development effort by **~70%**.
- Architected the system using microservices (**HLD + LLD**) with clearly defined service boundaries (Intelligence, Workspace, Account), enabling independent scaling, modular development, and efficient deployment of critical components.
- Implemented event-driven architecture using **Kafka** to handle asynchronous AI workflows (**file generation → storage → preview execution**), improving system responsiveness by **~60%** and ensuring reliable and fault-tolerant processing.
- Designed optimized data storage using **PostgreSQL** (metadata) and **MinIO** (object storage), applied indexing and query optimization strategies to reduce file retrieval latency by **~40%** and improve overall system performance.
- Built a dynamic preview infrastructure using **Kubernetes** where each project runs in isolated pods, and implemented **Redis**-based routing with a custom **reverse proxy** to map subdomains to ephemeral containers in real-time for scalable execution.
- Enforced production-ready coding standards using **Spring AI** with **Retrieval-Augmented Generation (RAG)**, and deployed applications on **GKE** using **GitHub Actions CI/CD** and **Qodana** for code quality, improving reliability and maintainability.

AI Mental Health Assistant | 24/7 Personal Mental Coach

[Demo](#) | [Code](#) | [Link](#)

- Designed and developed this system using **Angular and FastAPI**, enabling real-time conversational support for users.
- Implemented **NLP-based emotion detection pipeline**, classifying user inputs into emotional categories (e.g., stress, sadness) to drive context-aware response generation.
- Engineered a **personalization and memory system** by leveraging user interaction history in **PostgreSQL**, improving response relevance and increasing user engagement by **~30%**.
- Built an analytics dashboard using **Chart.js**, visualizing emotion distribution, user activity trends, and stress vs productivity correlations.
- Designed and analyzed a **feedback-driven** evaluation system, achieving **~68%** positive response rate and generating actionable insights on user behavior patterns.

CERTIFICATIONS

AWS Certified Developer - Associate

[View](#)

Spring Boot 0 to 100 Cohort 4.0 [AI + DevOps]

[View](#)

EDUCATIONS

Master of Business Administration in **Data Science**, Amity University, **CGPA: 8.42/10**.

July 2024 – June 2026

Bachelor of Technology in **Computer Science**, Rajiv Gandhi Proudयोगiki Vishwavidyalaya, **CGPA: 8.88/10**.

July 2019 – June 2023